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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program:	BSCS 1-1	**Date:**	SEPTEMBER 16, 2026

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just “programming.” As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If

you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, testing, and building applications and algorithms-driven software systems	Software Engineer, Backend Developer, Full-Stack Engineer
Systems & Cloud Infrastructure	Maintaining hardware provisioning, network virtualizing, uptime, and deployment pipelines	System Administrator, DevOps Engineer, Sote Reliability Engineer (SRE)
Data Science & AI Engineering	Extracting insights from structured/unstructured data and training predictive AI models	Data Scientist, Machine Learning Engineer, Data Engineer
Cybersecurity & Threat Defense	Safeguarding network perimeters, conducting penetration testing, and managing incident response	Information Security Analyst, Penetration Tester, SOC Specialist
Information Systems & Enterprise IT	Aligning business workflows with technology solution and enterprise software architectures	Business Systems Analyst, IT Manager, Solution Architect

My Chosen Pathway (Part B)

I am most interested in: Cyber Security & Threat Defense

Why I chose it:

I choose this pathway because I am interested in hacking and protecting others privacy also it's interesting to go to this pathway

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	• SOC Analyst (Tier 1) • Cybersecurity Specialist / Junior Analyst • Information Security Technician	• Monitoring SIEM alerts and analyzing network traffic. • Triage initial security events and perform basic log analysis. • Vulnerability scanning and patch management. • Enforcing basic access controls and password policies.
Mid-Level	• Incident Response Analyst • Cyber Threat Intelligence Analyst • Penetration Tester / Ethical Hacke	• Leading incident containment and forensic investigations. • Conducting penetration testing and threat hunting. • Analyzing malware and indicators of compromise (IoCs) . • Developing automated detection rules and response playbooks.
Senior-Level	• Cybersecurity Architect • Lead Threat Defense Engineer • Senior Information Security Manager	• Designing resilient network and cloud security architectures. • Implementing Zero Trust Architecture and defense-in-depth strategies. • Conducting risk assessments and regulatory compliance audits. • Managing security projects and mentoring junior analysts.

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Expert / Leadership	• Chief Information Security Officer (CISO) • VP of Cybersecurity / Director of Security Operations	• Setting enterprise-wide cybersecurity strategy and governance. • Managing cybersecurity budget, compliance, and legal risk. • Overseeing organizational cyber resilience and business continuity . • Communicating security posture to executive boards and key stakeholders

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Networking & Protocols	N	Watch YouTube tutorials and practice networking basics using online courses like TryHackMe.
Linux & Command Line Administration	N	Install Linux on my computer and practice terminal commands daily.
SIEM & Log Analysis	N	Learn basic log analysis using tools like Splunk and follow online step-by-step guides.
Threat Incidents and Incident response	N	Read about basic security incidents and practice solving entry-level scenarios on Let'sDefend.
Vulnerability & Assessment	N	Learn how to use basic scanning tools like Nmap trough simple online security lessons.
Python Scripting & Automation for Security	N	Watch free Python beginner courses to learn basic coding and writing simple scripts.

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Senior Cybersecurity Analyst
Company	Accenture
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	LinkedIn

Item	Details
Required Skills Listed (3–5)	- Advanced threat hunting, forensic analysis, and complex incident response handling. - Relevant senior certificate (e.g., CISSP, CISM, GCFA, or CEH) -Advance scripting for automation using Python, PowerShell, or Bash.

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I choose the Cyber Security because I like to protect digital system and also I want to expand my knowledge on how to defend against cyber threat.

2. Looking at your roadmap, which career stage excites you the most, and why?

The career that excite me the most, maybe becoming a senior level in the fields of cyber security because this is the phase I can handle and have a great knowledge about this field also to have a great skills in this field

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

I think the hardest skill for me to develop is mastering the SIEM log analysis and scripting, as it require deep technical knowledge and extensive hands on practice.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

Making this roadmap I think the cyber security is suited for me because I like how the cyber security work and also this career is suited for my character.

VII. Scoring Guide

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Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date